

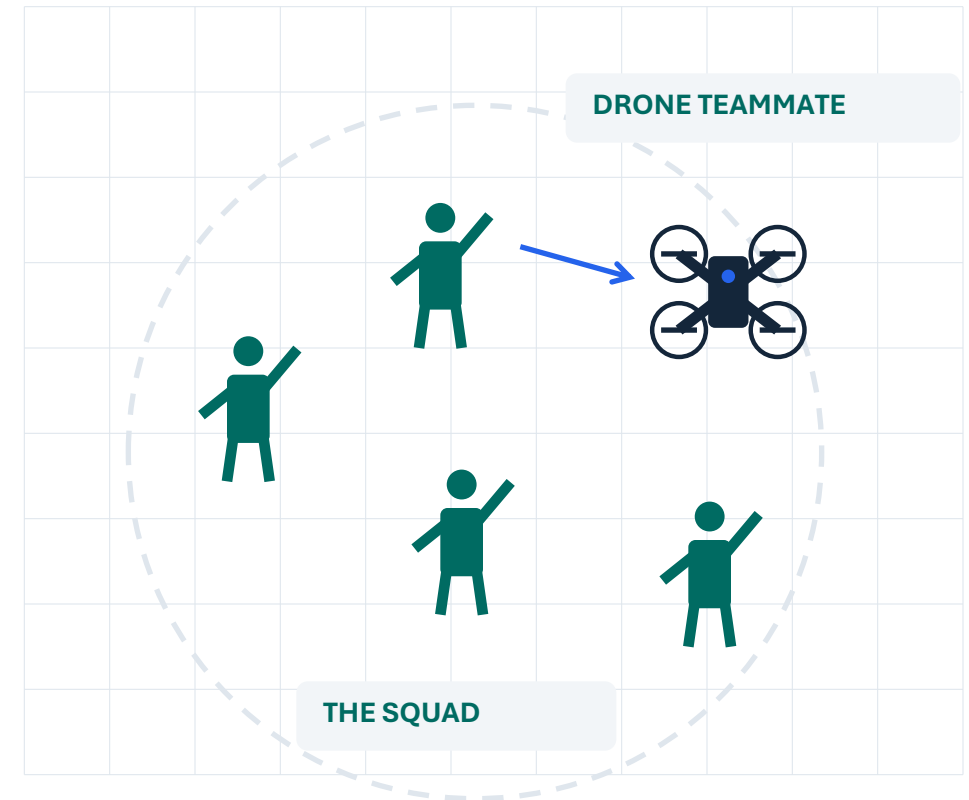
DEFENSE / WORKING SIMULATION

Add the drone to the squad.

Plan the mission. Signal changes as the squad moves.

ONE SQUAD. SHARED DRONE SUPPORT.

WHY NOW: Army drone specialists may join squads. [1]



ILLUSTRATIVE SQUAD CONCEPT

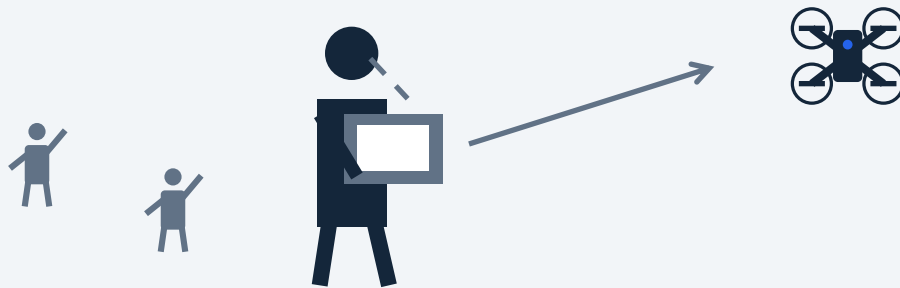
[1] Task & Purpose · Aug. 31, 2026

The squad needs to adapt. Eyes stay up.

OPERATING CONTEXT: GPS-DENIED ENVIRONMENTS

WORKFLOW DESCRIBED TO US

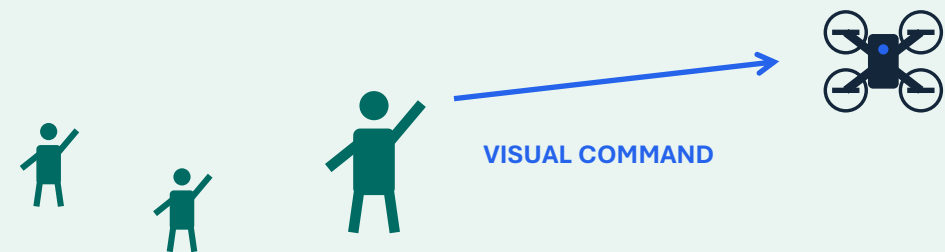
Small, chest-mounted tablet



Attention moves to the interface.

TACTLINK / ONBOARD VISION CONCEPT

Signal directly to the drone



Change intent while facing the squad.

INTENDED IMPACT

Redirect drone support as squad needs change.

Discovery: team conversation with an industry representative (paraphrased; informal user input).

Today: connected simulation. Onboard vision is planned; GPS-independent navigation is a separate requirement.

Human intent, in more than one form.

THE HACKATHON PROMPT

“Flight paths in plain English”

Plan and adjust a flight path using natural language.



OUR EXTENSION

Why not communicate with hand gestures, too?

Another human way to express intent and change drone behavior.

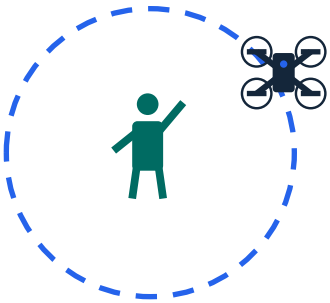
Natural-language planning inspired us. Gesture interaction became our working demonstration.

Source: DNHacks Defense inspiration prompt, “Flight paths in plain English” (provided by team).

Four commands. One continuous demo.

01

Orbit



Circle the operator.

02

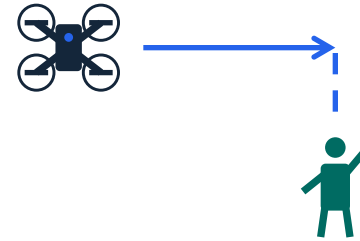
Move



Follow directional intent.

03

Return



Overhead to the sender.

04

Stop



Open-palm emergency stop.

LIVE SETUP

Phones as operators → Main computer → Shared 3D simulation

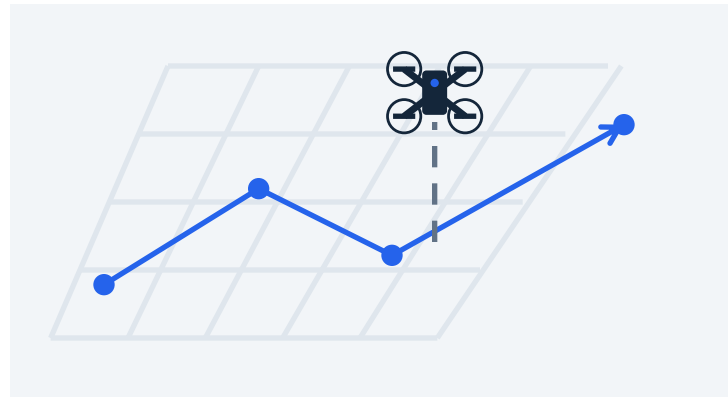
Plan the route. Adapt mid-flight.

A planned mission stays responsive to the operator.



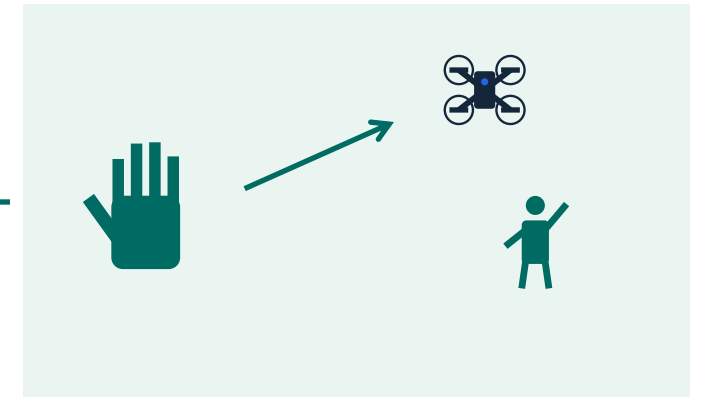
01 PLAN IN 2D

Build the mission in a flight planner.



02 VISUALIZE IN 3D

See the mission in a virtual world.



03 INTERRUPT BY GESTURE

Change drone behavior mid-flight.

POSSIBLE USE CASE

Send a drone between operators to provide overwatch.

Build the vocabulary around the team.

Custom gestures and natural-language commands give us a starting point.

- EXPAND** Train more custom signals with examples from different people.
- CONNECT** Bring gestures, spoken routes, and changes of plan into one interaction.
- REFINE** Make it clear which teammate is directing the drone and what it understood.

Test it with the people who will use it.

Turn a working demonstration into evidence we can build on.

01 / USER FEEDBACK

Learn from operators

Use feedback from industry representatives and prospective users to refine the commands.

02 / MEASUREMENT

Measure the interaction

Track recognition errors, response time, and unintended commands across repeatable trials.

03 / CHANGING CONDITIONS

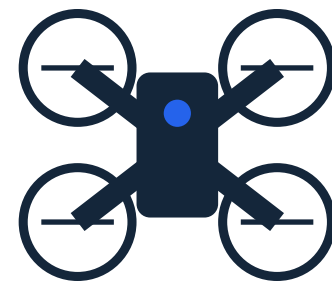
Test the difficult cases

Vary lighting, visibility, movement, and competing instructions from different teammates.

Bring the interaction onto the aircraft.

Talk to it. Gesture to it. Change the plan together.

Integrate onboard sensing and compute, connect the command system to flight control, and validate through controlled hardware trials.



THE GOAL: A drone that feels like another member of the team.